

Sportzfy: Comprehensive Requirement Analysis & Market Validation Report

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1. Executive Summary

This report validates the requirements for **Sportzfy**, an on-demand real-time turf booking engine. The analysis synthesizes **22 qualitative community screenshots** from local football groups and a **25-respondent quantitative market survey** from the CUET student and turf-player community.

2. Qualitative Community Evidence (Photos Analysis)

Evidence / Post	Real Problem Observed	Validated Feature
Mohammad Nur: "Ajk rat 8 tai ctg e kono turf khali hbe?"	Zero real-time visibility into open slots tonight.	Live Slot Availability Grid
Abdullah: "6 tarik rate 12am/1am 7v7 available ache?"	High demand for late night (12 AM-1 AM) and 7v7 format.	Time & Pitch Format (6v6/7v7) Filters
Anon: "any solid liable goalkeeper out there? match tomorrow"	Incomplete teams needing specific roles (especially GK).	Role-Based Matchmaking / GK Board
Saruar: "Dominatrix FC Team Match khelbaa?"	Amateur squads seeking rival teams for matches.	Team vs Team Matchmaking
Eco Sports: "Call or WhatsApp sms for booking..."	Turf owners overwhelmed with manual booking calls.	Turf Owner Self-Service Dashboard
Post: "Apollo turf e 7v7 khela jai? Slot price koto?"	Lack of transparent turf dimensions & pricing info.	Transparent Turf Spec Cards

3. Quantitative Survey Findings (25 Respondents)

Survey Dimension	Key Distribution / Findings	System Implication
Current Booking Method	60% Always Call, 16% Sometimes Call, 16% In-person (76% Manual)	Need automated live booking engine
Decision Priority	40% Availability/Timing, 32% Price, 24% Turf Quality	Prominent real-time filters & price display
Peak Playing Hours	52% Night, 36% Evening (88% Evening/Night concentrated)	Focus slot schedules on 6 PM - 1 AM window
Upfront Pay & Refund Policy	72% state Refund Policy is CRUCIAL before paying upfront	Automated transparent refund workflow
In-App Split Payment	76% Extremely Useful, 16% Somewhat Useful (92% positive)	Must support split-bill via payment links
Matchmaking Demand	48% Frequently seek teams, 32% Occasionally (80% Addressable)	Dedicated open match & player roster hub
Equipment Rental Add-on	60% Often forget/need gear, 20% need bibs (80% open)	Add gear selection checkboxes at checkout
1-Click Share Invites	76% Very Important, 20% Somewhat Important (96% demand)	1-Click WhatsApp / Messenger invite link
Must-Have Amenities	48% Clean washrooms/changing rooms, 32% Drinking water	Mandatory facility amenity filter tags
AI Weather Forecasts	60% state weather frequently ruins match plans	AI / automated rain alerts for booked slots

4. MoSCoW Feature Prioritization & Database Additions

- **Must Have:** Real-time slot locking engine (hold/release), bKash/Nagad gateway with automated refund workflow, Turf Owner dashboard, Multi-parameter Search & Filter (Time, Price, Format).
- **Should Have:** In-app split payment links, Matchmaking Hub (solo GK + team vs team), 1-click WhatsApp match invite sharing, Equipment rental

selector, Amenities filters.

- **Could Have:** AI weather rain warnings, Loyalty discount points, Dynamic pricing suggestions for owners.
- **Schema Additions:** *split_payment_status*, *equipment_rented* JSONB, *amenities* array, *pitch_formats* array, and *MatchRequests* table.

5. What Needs to Be Implemented & Technical Details

Module & Area	Technical Implementation Details
Module 1: Real-Time Discovery & Dynamic Filtering	• Interactive visual timetable grid showing hourly slots (■ Available, ■ Locked, ■ Booked). • Multi-filter bar: Zones (Halishahar, Chandgaon, Khulshi), Format (5v5, 6v6, 7v7), Time Windows (Evening 5-8 PM, Night 8 PM-1 AM), Amenities tags, and Price sliders. • Turf profile pages with verified reviews, pitch grass grade, and Google Maps embed.
Module 2: Concurrency Control & Instant Slot Lock (Hold & Release)	• 5-minute temporary lock in Redis/PostgreSQL upon clicking "Book Now" (TTL = 300s). • WebSocket channel (/ws/turfs/{id}/slots) broadcasting instant yellow lock to all concurrent users. • Background Celery worker automatically freeing unpaid slots back to green availability.
Module 3: Payment Gateway & Automated Refund Policy Engine	• Direct integration with bKash, Nagad, and Rocket mobile wallet sandbox APIs. • Rule-based refund automation: 100% refund (>12h before match), 50% refund (6-12h), 0% (<6h). • Severe weather rainout auto-refund mechanism + digital booking QR code receipt.
Module 4: In-App Split Payment & 1-Click WhatsApp Invite Sharing	• Bill-split calculator dividing slot total into per-player payment sub-links. • 1-Click WhatsApp invite generator formatting date, time, venue Google Maps link, and split link. • Real-time captain dashboard tracking which teammates have completed their split payment.
Module 5: Matchmaking Hub & Role-Based Player Recruitment	• Team vs Team (Rival match) board: Booked teams post open challenges to share turf fees. • Solo Recruitment board: Teams recruit specific roles (e.g. Goalkeepers) with cost share & timing. • Player profiles (Role, Skill rating) with instant join requests and WhatsApp alerts.
Module 6: Equipment Rental Selector (Digital POS)	• Checkout checkboxes for supplementary gear (Match Ball +150 BDT, Bibs +100 BDT, Gloves +100 BDT). • Turf owners manage gear inventory and pricing directly from their portal.
Module 7: Turf Owner & Platform Admin Portals	• Turf owners manage pitch formats, upload photos, set peak/off-peak pricing, and log manual walk-ins. • Admin panel for turf verification, user moderation, and platform revenue settlement.
Module 8: AI Weather Alert & Intelligent Recommendations	• Geolocation-based rain probability integration via OpenWeatherMap API with 3h pre-match alerts. • Recommendation model analyzing user booking history to suggest optimal time slots and venues.